NOTE: *Depending on market region, vehicle configuration, options purchased, Self-Driving hardware, and software version, your vehicle may not be equipped with Full Self-Driving (Supervised), or the feature may not operate exactly as described.*

When Full Self-Driving (Supervised) is engaged, Model Y attempts to drive to your destination by following curves in the road, stopping at and negotiating intersections, making left and right turns, navigating roundabouts, and entering/exiting highways.

Unlike Traffic-Aware Cruise Control, Autosteer, and Navigate on Autosteer, which are intended for use on multi-lane roadways with clear lane markings, Full Self-Driving (Supervised) is meant to work in a variety of driving scenarios. You can use Full Self-Driving (Supervised) on any type of roadway, including residential and city streets.

WARNING: Driver intervention may be required in certain situations, such as on narrow roads with oncoming cars, in construction zones, or while going through complex intersections. For more examples of scenarios in which driver intervention might be required, see Limitations and Warnings on page 136.

Full Self-Driving (Supervised) uses inputs from cameras mounted at the front, rear, left, and right of Model Y to build a model of the area surrounding Model Y (see Cameras on page 20). The AI computer installed in Model Y is designed to use this input, rapidly process neural networks, and make decisions to guide you to your destination.

NOTE: As Tesla's Full Self-Driving (Supervised) capabilities evolve, Model Y is upgraded through over-the-air software updates. Download updates as soon as they become available.

Like other Self-Driving features, Full Self-Driving (Supervised) requires a fully attentive driver and will display a series of escalating warnings requiring driver response. **You must remain attentive and be ready to take over at all times while Full Self-Driving (Supervised) is engaged.** While Full Self-Driving (Supervised) is engaged, the cabin camera monitors driver attentiveness (see Driver Attentiveness on page 108).

WARNING: Full Self-Driving (Supervised) requires you to pay attention to the road and be ready to take over at all times. Remain attentive, be mindful of road conditions and surrounding traffic, pay attention to pedestrians and cyclists, and always be prepared to take immediate action (especially around blind corners, crossing intersections, and in narrow driving situations). Failure to follow these instructions could cause damage, serious injury or death. It is your responsibility to familiarize yourself with the limitations of Full Self-Driving (Supervised) and the situations in which it may not work as expected. For more information, see Self-Driving Limitations and Warnings on page 136.

CAUTION: Use of Full Self-Driving (Supervised) will be suspended if improper usage is detected. For more information, see Self-Driving Suspension on page 109.

CAUTION: As Full Self-Driving (Supervised) deployment expands to additional countries, it is essential for drivers using Full Self-Driving (Supervised) in newly eligible countries to be extra attentive and overly cautious. Every country contains unique infrastructure, driving behaviors, and traffic patterns that Full Self-Driving (Supervised) must adapt to over time. You must be ready to take over safely at any time.

Before Using Full Self-Driving (Supervised)

1. Enable Full Self-Driving (Supervised). While Model Y is parked, touch **Controls** > **Self-Driving** > **Full Self-Driving (Supervised)** and then, after carefully reading and understanding the popup window, touch **Yes**.

   NOTE: Model Y must be in Park for you to enable Full Self-Driving (Supervised), or to disable it and instead enable Autosteer or Traffic-Aware Cruise Control.

2. (Optional) Customize Full Self-Driving (Supervised) settings.
   - **Speed Profile**: If desired, change the default setting of **Sloth**. For more information, see Choosing a Speed Profile on page 105
   - **Use HOV Lanes**: Choose whether Full Self-Driving (Supervised) uses High Occupancy Vehicle (HOV) lanes on your route.

3. Ensure that the cameras on Model Y are unobstructed and calibrated (see Cameras on page 20). Full Self-Driving (Supervised) depends on the ability of the cameras to detect traffic lights, stop signs, and road markings.

   NOTE: Before

   WARNING: Do not use Full Self-Driving (Supervised) if anything, such as a ball hitch, bike rack, or trailer, is attached to the rear of your vehicle. If Model Y detects that a camera is blocked (for example, by a rear-mounted bike rack), the touchscreen displays an alert and Full Self-Driving (Supervised) performance may be degraded or, in some cases, the feature is unavailable.

4. Ensure that the latest version of maps has been downloaded to Model Y (see Map Updates on page 175). Although Full Self-Driving (Supervised) primarily uses visual data received from the vehicle's cameras, greater accuracy is achieved when using the most recent map data.

Choosing a Speed Profile

Choose a Speed profile to determine how quickly Model Y drives while Full Self-Driving (Supervised) is engaged, and how frequently it makes speed-based lane changes. In addition to the selected speed profile, Model Y uses navigation data, historical data from other Tesla vehicles, the speed of other vehicles on the road, and posted speed limits to determine how quickly to drive.

The default speed profile is **Sloth**. To change the speed profile:

- Tap **Self-Driving** on the touchscreen while Full Self-Driving (Supervised) is engaged.
- Touch **Controls** > **Self-Driving**.
- Roll the right scroll button on the steering wheel up or down.

Once you select a speed profile, it is selected by default the next time you use Full Self-Driving (Supervised). The selected speed profile is also associated with your driver profile (see Driver Profiles on page 94).

Sloth: Model Y drives at or below the speed limit.

Chill: Model Y drives in slower lanes and makes minimal lane changes.

Standard: Model Y drives at normal speed, and adjusts speed to match the flow of traffic.

Hurry: Model Y drive faster and makes more frequent lane changes.

Mad Max: Model Y drives more quickly than in the **Hurry** profile, with more frequent lane changes.

Starting Full Self-Driving (Supervised)

As long as Full Self-Driving (Supervised) is enabled and available, you can activate it at any speed less than 85 mph (140 km/h) (including when Model Y is at a standstill) by pressing the right scroll button.

NOTE: To engage Full Self-Driving (Supervised), headlights must be set to **On** or **Auto**. Although Full Self-Driving (Supervised) is available both during the day and in low light conditions (dusk or dark), it aborts or is unavailable if headlights are set to **Off**. When Full Self-Driving (Supervised) is engaged, **Adaptive Headlights** (or, on some vehicles, **Auto High Beam**) is automatically enabled (see High Beam Headlights on page 81) and wipers are set to **Auto**.

When Full Self-Driving (Supervised) is engaged, **Self-Driving** is shown in blue on the touchscreen, accompanied by the Full Self-Driving (Supervised) profile (see Choosing a Speed Profile on page 105).

If you do not choose a destination, Model Y chooses the most probable driving path or suggests a destination based on your driving.

NOTE: You can override Full Self-Driving (Supervised) at any time if you are uncomfortable or uncertain about the system's intended course of action. You are driving Model Y at all times.

Starting Full Self-Driving (Supervised) from the Touchscreen

You can also activate Full Self-Driving (Supervised) from the touchscreen, either when Model Y is in Drive or in Park.

1. First, enable this feature by touching **Controls** > **Tap to Start FSD (Supervised)**.

   Brake Confirm is enabled by default. When **Brake Confirm** is enabled, you will need to briefly press the brake pedal to confirm each time you start Full Self-Driving (Supervised) from Park.

2. (Optional) Enter a destination. If you do not choose a destination, Model Y chooses the most probable driving path or suggests a destination based on your driving.
3. Touch **Start Self-Driving** on the touchscreen and, if prompted, press the brake pedal to continue. If Model Y is in Park, the vehicle shifts into Drive or Reverse, pulls out of the parking space, and begins navigating to your destination. If Model Y is in Drive, Full Self-Driving (Supervised) engages immediately and navigates Model Y to the set destination.

To cancel Full Self-Driving (Supervised) once Model Y has begun the trip, tap the brake pedal or use any of the methods listed in Canceling Full Self-Driving (Supervised) on page 108.

CAUTION: Start FSD (Supervised) from Park is available as long as Full Self-Driving (Supervised) is enabled and available and Model Y detects that a suitable driver is present. Start FSD (Supervised) from Park may be unavailable for a number of reasons, including (but not limited to):

- Model Y detects that there is no suitable driver in the driver's seat.
- Model Y detects that the driver's seat belt is not buckled.
- A door, the trunk, or the frunk is open.
- A charge cable is connected to the vehicle.
- Model Y is in Valet Mode, Track Mode, or any other mode that restricts Self-Driving usage.

CAUTION: The use of devices designed to circumvent occupancy detection is a violation of the terms of use for Full Self-Driving (Supervised) and may result in the feature being permanently disabled on your vehicle and a ban on future use of the feature.

While Using Full Self-Driving (Supervised)

Full Self-Driving (Supervised) changes lanes, makes left and right turns, follows on- and off-ramps, and takes forks in the road as necessary to reach the destination.

WARNING: NEVER make assumptions and predict when and where Full Self-Driving (Supervised) will stop or continue through an intersection or road marking. From a driver's perspective, the behavior of Full Self-Driving (Supervised) may appear inconsistent. Always pay attention to the roadway and be prepared to take immediate action. It is the driver's responsibility to determine whether to stop or continue through an intersection. Never depend on Full Self-Driving (Supervised) to determine when it is safe and/or appropriate to stop or continue through an intersection.

The touchscreen displays a visualization of the environment surrounding Model Y, including the roadway and detected objects such as vehicles, pedestrians, curbs, bicyclists, and more. Objects that are highlighted on the visualization represent priorities that Full Self-Driving (Supervised) is actively controlling for at a given time.

NOTE: The Full Self-Driving (Supervised) visualization may not be a holistic representation of the objects, road markings, road signals, and other variables that Full Self-Driving (Supervised) takes into account as it attempts to drive to your destination. While Full Self-Driving (Supervised) is engaged, it uses data from the cameras on Model Y that may not be represented in the visualization (see Cameras on page 20).

NOTE: You can widen the visualization area to fullscreen. Enable the setting by touching **Controls** > **Self-Driving** and then touching **Expanded Full Self-Driving Visualization**. Then on the touchscreen, swipe the visualization window handle to fill the entire screen.

Like Autosteer and Traffic-Aware Cruise Control, Full Self-Driving (Supervised) maintains your speed and following distance from the vehicle ahead of Model Y, if there is one. Full Self-Driving (Supervised) also slows down and stops at traffic lights and stop signs as necessary, and reacts to pedestrians, cyclists, and other vehicles on the road.

For example, if you are driving on a residential street and another vehicle backs out of a driveway ahead of Model Y, Full Self-Driving (Supervised) slows down or stops as appropriate. If the other vehicle stops backing out while partially blocking the driving lane, Full Self-Driving (Supervised) slows down and maneuvers around the other vehicle if the width of the lane allows it.

When the traffic in front of you is slowing down, Full Self-Driving (Supervised) shows blue arrows in the driving lane and slows down to maintain an appropriate following distance from the vehicle ahead of you. When you are driving on the highway, Full Self-Driving (Supervised) displays a message on the touchscreen to inform you when an action (such as a lane change) is being taken.

WARNING: In rare cases, Full Self-Driving (Supervised) may not appropriately slow down, come to a stop, or resume control for a stop sign or traffic light. You may assist the system by lightly applying the accelerator, or can override Full Self-Driving (Supervised) at any time.

Changing Lanes

Full Self-Driving (Supervised) initiates its own lane changes and turns based on the driving environment. Unlike Navigate on Autosteer, Full Self-Driving (Supervised) does not require confirmation before a lane change. You can also indicate your preference to make a lane change or turn by using the right or left turn signal, respectively.

WARNING: When a lane change or turn is in progress, turn signal inputs may not cancel the maneuver. To cancel a lane change or turn, intervene with the steering wheel or other vehicle controls (see #unique_352 on page).

Take Over Immediately

In situations where Self-Driving is unable to steer Model Y, a warning chime sounds and the touchscreen displays the following message.

Take over immediately

When you see this message, **take over steering immediately**.

Arriving at Your Destination

You can choose what Model Y does when you reach your destination. While Full Self-Driving (Supervised) is engaged, the destination behavior (if there is one selected) is displayed on the touchscreen.

To choose the destination behavior, touch **Self-Driving** in the car status section of the touchscreen. Under **At Destination**, choose from the following options.

- **Park**: If a suitable parking space is detected at your destination, Full Self-Driving (Supervised) parks Model Y.

  NOTE: If no suitable parking space is detected, Model Y pulls over instead.

- **Curbside**: Once Model Y reaches your destination, the vehicle pulls over for a short term stop (to pick up or drop off a passenger, for example).

If you selected **Park**, you can also select what kind of parking space you would like Model Y to choose, such as **Parking Lot**, **Street**, or **Driveway**. The available options depend on your destination. If you are navigating to a Supercharger, for example, you can select **Charger**.

When you choose an option, Model Y defaults to the same behavior the next time you use Full Self-Driving (Supervised) to navigate to the same destination.

In either case, once navigation is complete Model Y shifts into Park.

Canceling Full Self-Driving (Supervised)

To disengage Full Self-Driving (Supervised), do any of the following:

- Press the brake pedal.
- Press the right scroll button on the steering wheel.
- Take over and steer manually.

In addition, Full Self-Driving (Supervised) will disengage if any of the following occurs:

- You shift out of Drive.
- A door or the trunk is opened.
- There is an Automatic Emergency Braking event (see Collision Avoidance Assist on page 143).
- The driver's seat belt is released, and/or the driver gets out of their seat.
- You do not respond to repeated reminders to pay attention and subsequent messages on the touchscreen.
- Full Self-Driving (Supervised) becomes unavailable. This can happen for a number of reasons (for example, if a camera becomes obscured). If Full Self-Driving (Supervised) disengages, an alert will appear on the touchscreen to notify you and a chime will sound. If this happens, **take control of steering immediately**.

When driver intervention is required, it is best to safely disengage as soon as possible. It is recommended to practice disengaging from Full Self-Driving (Supervised) in a safe environment without other road users so you may become familiar with the process.

Driver Attentiveness

Full Self-Driving (Supervised) requires that the driver pay attention to the road, their surroundings, and other road users.

The cabin camera monitors continued driver attentiveness when Full Self-Driving (Supervised) is engaged.

When the cabin camera is actively monitoring driver attentiveness, a green indicator appears in the status bar.

This system cannot be disabled. The cabin camera does not require full visibility of the driver's eyes in order to monitor attentiveness. The system is still active, for example, if the driver is wearing sunglasses.

If the driver repeatedly looks away from the road, the touchscreen displays a warning. The warning is dismissed once the driver begins paying attention to the road again.

Please pay attention to the road

If the cabin camera does not have clear visibility of the driver's hand and arm locations, the touchscreen periodically displays a message reminding the driver to apply slight force to the steering wheel.

Apply slight turning force to steering wheel

If the driver repeatedly ignore prompts to apply slight force to the steering wheel or to pay attention, Self-Driving displays a series of escalating warnings and, if those warnings are ignored, disables for the rest of the drive and displays the following message.

Self-Driving temporarily disabled
Driver attention warnings ignored

If the driver does not resume manual steering, Model Y sounds a continuous chime, turns on the warning flashers, and slows to a complete stop.

WARNING: Do not use handheld devices while using Self-Driving features. If the cabin camera detects a handheld device while Self-Driving is engaged, the touchscreen displays a message reminding you to pay attention.

WARNING: The use of devices designed to circumvent driver attentiveness is a violation of the terms of use for Self-Driving and may result in the feature being permanently disabled on your vehicle and a ban on future use of the feature.

Self-Driving Suspension

Use of Self-Driving features will be suspended if improper usage is detected.

Use of Autosteer (if equipped) and Full Self-Driving (Supervised) is suspended for a week when you or another driver of your vehicle receives five "strikeouts." A strikeout is when the Self-Driving system disengages for the remainder of a trip after the driver receives several audio and visual warnings for inattentiveness.

You can see how many strikeouts are remaining before Self-Driving access is suspended by touching **Controls** > **Self-Driving**.

A strikeout is forgiven after half a week (3.5 days), as long as you don't receive another strikeout in that time.

NOTE: If your access to Autosteer (if equipped) and Full Self-Driving (Supervised) is suspended, you also lose access to Autopark. However, you can still use Traffic-Aware Cruise Control and all active safety features are still enabled.

There may be occasions where driver intervention is required and you must take over immediately to maintain safe driving. Driver-initiated disengagements do not count as improper usage and are expected from the driver.

Full Self-Driving (Supervised) Limitations

Warnings and Cautions

WARNING: Always remember that Full Self-Driving (Supervised) (also known as Autosteer on City Streets) does not make Model Y autonomous and requires a fully attentive driver who is ready to take immediate action at all times.

WARNING: Full Self-Driving (Supervised) requires you to pay attention to the road and be ready to take over at all times. Remain attentive, be mindful of road conditions and surrounding traffic, pay attention to pedestrians and cyclists, and always be prepared to take immediate action (especially around blind corners, crossing intersections, and in narrow driving situations). Failure to follow these instructions could cause damage, serious injury or death. It is your responsibility to familiarize yourself with the limitations of Full Self-Driving (Supervised) and the situations in which it may not work as expected.

WARNING: Failure to follow all warnings and instructions can result in property damage, serious injury or death.

CAUTION: Full Self-Driving (Supervised) and its associated functions may not operate as intended and there are numerous situations in which driver intervention may be needed. Examples include (but are not limited to):

- Interactions with pedestrians, bicyclists, and other road users.
- Unprotected turns with high-speed cross traffic.
- Multi-lane turns.
- Simultaneous lane changes.
- Narrow roads with oncoming cars or double-parked vehicles.
- Rare objects such as trailers, ramps, cargo, open doors, etc. protruding from vehicles.
- Merges onto high-traffic, high-speed roads.
- Debris in the road.
- Construction zones.
- High curvature roads, particularly at fast driving speeds.

Visibility is critical for Full Self-Driving (Supervised) to operate. Low visibility, such as low light or poor weather conditions (rain, snow, direct sun, fog, etc.) can significantly degrade performance.

WARNING: Model Y may quickly and suddenly make unexpected maneuvers or mistakes that require immediate driver intervention.

The list above represents only a fraction of the possible scenarios that can cause Full Self-Driving (Supervised) to make sudden maneuvers and behave unexpectedly. In fact, Model Y can suddenly swerve even when driving conditions appear normal and straight-forward. Stay alert and always pay attention to the roadway so you can anticipate the need to take corrective action as early as possible. Remember that this is an early access feature that must be used with extra caution.

CAUTION: As Full Self-Driving (Supervised) deployment expands, Tesla will gradually make it available to eligible customers in select countries outside the United States. Because every country contains unique infrastructure, driving behaviors, and traffic patterns that Full Self-Driving (Supervised) must adapt to over time, it is essential for drivers using Full Self-Driving (Supervised) in newly eligible countries to be extra attentive and overly cautious. You must be ready to take over safely at any time.

About Self-Driving

Self-Driving refers to a suite of advanced driver assistance features that are intended to make driving more convenient and less stressful. None of these features make Model Y fully autonomous or replace you as the driver.

It is your responsibility to familiarize yourself with the limitations of these features, pay attention to the road, and be ready to take immediate action at any time. For more limitations, cautions, and warnings, see Self-Driving Limitations and Warnings on page 136.

Traffic-Aware Cruise Control

All Tesla vehicles are equipped with Traffic-Aware Cruise Control. Traffic-Aware Cruise Control maintains your speed and an adjustable following distance from the vehicle in front of you, if there is one. For more information, see Traffic-Aware Cruise Control on page 112.

Full Self-Driving (Supervised)

Full Self-Driving (Supervised) attempts to drive to your destination by following curves in the road; stopping at and negotiating intersections, stop signs, and roundabouts; making left and right turns; and entering/exiting highways. For more information, see Full Self-Driving (Supervised) on page 105.

NOTE: *Depending on market region, vehicle configuration, options purchased, Self-Driving hardware, and software version, your vehicle may not be equipped with Full Self-Driving (Supervised), or the feature may not operate exactly as described.*

Other Self-Driving Features

In addition, the following Self-Driving features may be available for your vehicle. The specific features enabled depend on market region, date of manufacture, software version, Self-Driving hardware, and vehicle configuration. For information that is customized to your vehicle, view the Owner's Manual on your vehicle's touchscreen by touching the app launcher and then selecting the Manual app.

- **Autosteer**: Maintains your speed and distance from a leading vehicle while also intelligently keeping Model Y in its lane (see Autosteer on page 115). Autosteer includes **Auto Lane Change** (Auto Lane Change on page 116) and **Navigate on Autosteer** (see Navigate on Autosteer on page 117).

  NOTE: *Autosteer, Auto Lane Change, and Navigate on Autosteer are available for select Tesla vehicles. To check if your vehicle is equipped with a specific Self-Driving feature, refer to the Owner's Manual on your vehicle's touchscreen or touch **Controls** > **Self-Driving**. Feature availability depends on vehicle configuration, region, model, vehicle trim and model year.*

- **Autopark**: Parks Model Y, either parallel or perpendicularly (see Autopark on page 130).
- **Summon**: Allows you to park and retrieve Model Y using the Tesla mobile app while you are standing outside of your vehicle (see Summon on page 132).
- **Traffic Light and Stop Sign Control**: Maintains your speed, keeps a following distance, and keeps Model Y in its lane while also slowing down and stopping for traffic lights and stop signs (see Traffic Light and Stop Sign Control on page 123).

  NOTE: *Traffic Light and Stop Sign Control is available for select Tesla vehicles. To check if your vehicle is equipped with a specific Self-Driving feature, refer to the Owner's Manual on your vehicle's touchscreen or touch **Controls** > **Self-Driving**. Feature availability depends on vehicle configuration, region, model, vehicle trim and model year.*

How It Works

Self-Driving feature use the cameras on Model Y. There are cameras mounted on the front, rear, left, and right sides of Model Y (see Cameras on page 20).

Model Y is equipped with a cabin camera, mounted in the rear-view mirror, that monitors driver attentiveness. It is your responsibility to pay attention to the road and be ready to take immediate action at any time.

When Autosteer or Full Self-Driving (Supervised) is engaged, Model Y shows a series of escalating warnings reminding you of your responsibilities. If there is no response, Self-Driving disengages and is unavailable for the remainder of the drive.

WARNING: Self-Driving features do not guarantee collision warning or avoidance. It is your responsibility to stay alert, drive safely, and be in control of the vehicle at all times. Never depend on Self-Driving to adequately slow down Model Y. Always watch the road in front of you and be prepared to take corrective action at all times. Failure to do so can result in serious injury or death.

It is your responsibility to familiarize yourself with the limitations of Self-Driving and be ready to take control at all times. For more limitations, cautions, and warnings, see Self-Driving Limitations and Warnings on page 136.

Traffic-Aware Cruise Control

Traffic-Aware Cruise Control maintains your speed and an adjustable following distance from the vehicle in front of you, if there is one.

Before Using Traffic-Aware Cruise Control

Before you use Traffic-Aware Cruise Control, customize how it works by touching **Controls** > **Self-Driving**.

- **Set Speed**: Choose whether Traffic-Aware Cruise Control engages at the currently detected speed limit or your current driving speed. Touch **Controls** > **Self-Driving** and choose either **Speed Limit** or **Current Speed**.
- **Offset**: If you choose **Speed Limit**, you can specify an offset by touching **Set Speed Offset**. You can choose **Fixed** (the cruising speed adjusts by a specific amount on all roads) or **Percentage** (the cruising speed is adjusted as a percentage of the road's detected speed limit).
- **Green Traffic Light Chime**: In Canada and U.S.: If on, a chime will sound when you are waiting at a red traffic light and the light turns green. If you are not actively using Traffic-Aware Cruise Control and are waiting at a red light with a car in front of you, the chime sounds when the car ahead of you advances.

Using Traffic-Aware Cruise Control

To enable Traffic-Aware Cruise Control, touch **Controls** > **Self-Driving** > **Traffic-Aware Cruise Control**.

Traffic-Aware Cruise Control is available as long as you are driving between 18 mph (30 km/h) and 85 mph (140 km/h). You can activate Traffic-Aware Cruise Control at lower speeds if there is a vehicle detected at least 5 feet (1.5 meters) ahead of Model Y.

NOTE: Traffic-Aware Cruise Control is unavailable when Full Self-Driving (Supervised) is enabled (see Full Self-Driving (Supervised) on page 105).

To use Traffic-Aware Cruise Control:

1. Press the right scroll button, then release the accelerator pedal to allow Traffic-Aware Cruise Control to maintain the cruising speed. A chime sounds to indicate that Traffic-Aware Cruise Control is now active.
2. To change the set speed, roll the right scroll wheel up to increase, or down to decrease. You can apply the accelerator at any time to temporarily override the set cruising speed.
3. To cancel Traffic-Aware Cruise Control, press the right scroll button or press the brake pedal.

40 MAX

When Traffic-Aware Cruise Control is available but not engaged, the touchscreen displays the cruising speed in gray. The number shown represents the speed that will be set when you engage Traffic-Aware Cruise Control.

40 MAX

When Traffic-Aware Cruise Control is actively cruising at a set speed, the speed is highlighted with blue text.

WARNING: Traffic-Aware Cruise Control is designed for your driving comfort and convenience and is not a collision warning or avoidance system. It is your responsibility to stay alert, drive safely, and be in control of the vehicle at all times. Never depend on Traffic-Aware Cruise Control to adequately slow down Model Y. Always watch the road in front of you and be prepared to take corrective action at all times. Failure to do so can result in serious injury or death. For more information, see Self-Driving Limitations and Warnings on page 136.

While Using Traffic-Aware Cruise Control

When Traffic-Aware Cruise Control is active and maintaining a set speed, the speed is highlighted with blue text on the touchscreen.

Maintaining the Set Speed

When Traffic-Aware Cruise Control is active, Model Y maintains your set cruising speed whenever a vehicle is not detected in front of it. When cruising behind a vehicle, Model Y accelerates and decelerates as needed to maintain a chosen following distance, up to the set speed.

You can manually accelerate at any time by pressing the accelerator pedal, but when you release the pedal Model Y resumes cruising at the set speed.

Model Y also adjusts the cruising speed when entering and exiting curves.

When Model Y is actively slowing down to maintain the selected distance from the vehicle ahead, brake lights turn on. You may notice slight movement of the brake pedal. However, when Model Y is accelerating, the accelerator pedal does not move.

Changing the Set Speed

Roll the right scroll wheel up to increase, or down to decrease, the set speed.

To change the cruising speed to the current speed limit (including any offset you've specified), press the speed limit sign on the touchscreen.

It may take a few seconds for Model Y to reach the new cruising speed.

Adjusting the Following Distance

To adjust the following distance you want to maintain between Model Y and a vehicle traveling ahead of you, press the steering wheel's right scroll button to the left or right.

The closest following distance is 2.

Each setting corresponds to a time-based distance that represents how long it takes for Model Y, from its current location, to reach the location of the rear bumper of the vehicle ahead of you. Traffic-Aware Cruise Control retains your setting until you change it again.

As you adjust the following distance, the touchscreen displays the current setting.

Stopping and Slowdowns

When moving significantly faster than vehicles in adjacent lanes, Model Y automatically reduces the driving speed. This is especially helpful in heavy traffic situations or when vehicles are constantly merging into different lanes. When Model Y detects other vehicles driving significantly slower, the touchscreen highlights the adjacent lanes with arrows and detected vehicles in gray, and Model Y reduces the driving speed as appropriate. To temporarily override this feature, press the accelerator pedal.

When following a vehicle, Traffic-Aware Cruise Control remains active at low speeds, even when Model Y comes to a full stop. For example, Traffic-Aware Cruise Control remains active even if Model Y slows down to a complete or near-complete stop in heavy, stop-and-go traffic on a highway. When traffic starts moving more rapidly, Traffic-Aware Cruise Control again accelerates up to the set speed.

Sometimes when Model Y is at a full stop, Traffic-Aware Cruise Control goes into a HOLD state. If this happens, briefly press the accelerator pedal to resume cruising.

When the HOLD status is active, the touchscreen displays the HOLD icon and a message that indicates that you need to resume cruise control.

Model Y goes into HOLD state while Traffic-Aware Cruise Control is active in the following circumstances:

- Model Y has been at a standstill for 5 minutes.
- Model Y detects a pedestrian (the HOLD state may clear when the pedestrian is no longer detected).

- Model Y suddenly loses visibility of the vehicle in front of you.
- An obstacle is detected in front of Model Y.

Cruising Near or On Exits

When you are cruising near an exit on a controlled-access highway and engage the turn signal toward the off-ramp, Traffic-Aware Cruise Control assumes you are exiting and begins to slow down Model Y. If you do not drive onto the off-ramp, Traffic-Aware Cruise Control resumes cruising at the set speed.

In a region with right hand traffic, this occurs only when you engage the right turn signal when driving in the right-most lane within 164 ft. (50 meters) of an exit. Likewise in regions with left hand traffic, this occurs when engaging the left turn signal when driving in the left-most lane within 164 ft. (50 meters) of an exit.

When enabled while on a highway interchange or off-ramp, Traffic-Aware Cruise Control may reduce your set speed in 5 mph (5 km/h) increments – to as slow as 25 mph (40 km/h) – to better match the reported speeds of other Tesla vehicles that have driven at that specific location. To override this and continue cruising at your set speed, tap the accelerator pedal. The new set speed is maintained for the duration of the interchange or off-ramp (unless you override it or cancel Traffic-Aware Cruise Control). After the interchange or off-ramp, the set speed may revert or change as necessary based on the new location. For example, if you merged onto a different highway, the set cruising speed reverts to what it was before driving on the interchange.

WARNING: In some cases (such as having insufficient data), Traffic-Aware Cruise Control may not automatically reduce the set speed on the highway interchange or off-ramp. Do not rely on Traffic-Aware Cruise Control to determine an appropriate driving speed. Tesla recommends driving at a speed that is safe for road conditions and within posted speed limits.

When cruising onto an on-ramp to a controlled-access highway, Traffic-Aware Cruise Control automatically adjusts the set cruising speed to the speed limit of the highway, plus any offset you have specified.

Overtake Acceleration

Engage the turn signal momentarily to accelerate Model Y towards the vehicle ahead of it. By momentarily holding the turn signal, you can quickly accelerate up to your set speed without having to press the accelerator pedal as long as:

- Traffic-Aware Cruise Control is operating and detects a vehicle in front of you.
- No obstacles or vehicles are detected in the target lane.
- Model Y is traveling below the set speed, but over 45 mph (72 km/h).

Model Y stops accelerating when you reach your set cruising speed, if changing lanes takes too long, or if Model Y gets too close the vehicle ahead. Model Y also stops accelerating if you disengage the turn signal.

Stop Light and Stop Sign Warning

While Traffic-Aware Cruise Control is in use, Model Y displays a warning on the touchscreen and sounds a chime if it detects that you are likely to run through a red stop light or stop sign. If this happens, **TAKE IMMEDIATE CORRECTIVE ACTION!**

The visual and audible warnings cancel after a few seconds or when you press the brake pedal, whichever comes first.

Stop Light and Stop Sign Warning provides warnings only. It does not slow down or stop Model Y at red traffic lights, stop signs, road markings, etc. If equipped with Traffic Light and Stop Sign Control, you can enable this feature to automatically stop Model Y at traffic lights and stop signs (see Traffic Light and Stop Sign Control on page 123).

Canceling Traffic-Aware Cruise Control

Traffic-Aware Cruise Control cancels when:

- You press the right scroll button on the steering wheel.
- You press the brake pedal.
- You exceed 85 mph (140 km/h).
- You shift into Reverse, Park, or Neutral.
- A door is opened.
- An Automatic Emergency Braking event occurs (see Collision Avoidance Assist on page 143).
- The driver's seatbelt is released, and/or the driver gets out of their seat.

When Traffic-Aware Cruise Control cancels, the cruising speed icon on the touchscreen turns gray to indicate that Traffic-Aware Cruise Control is no longer active.

When Traffic-Aware Cruise Control cancels, Model Y does not coast. Instead, regenerative braking slows down Model Y in the same way as when you move your foot off the accelerator when driving without Traffic-Aware Cruise Control (see Regenerative Braking on page 86).

This topic describes how to enable and use the following driver assistance features.

NOTE: *Depending on options purchased, date of manufacture, software version, vehicle configuration, and Self-Driving hardware, your vehicle may not be equipped with the following features, or a feature may not operate exactly as described. Autosteer, Auto Lane Change, and Navigate on Autosteer are available for select Tesla vehicles. To check if your vehicle is equipped with a specific Self-Driving feature, refer to the Owner's Manual on your vehicle's touchscreen or touch* ***Controls*** *>* ***Self-Driving****. Feature availability depends on vehicle configuration, region, model, vehicle trim and model year.*

- **Autosteer**: Like Traffic-Aware Cruise Control, Autosteer maintains a set speed (if there is not a vehicle in front of you) or a set following distance (if there is a vehicle in front of you). In addition, Autosteer detects lane markings, road edges, and the presence of vehicles and objects to intelligently keep Model Y in its driving lane (see Autosteer on page 115).

  NOTE: Autosteer is a BETA feature.

- **Auto Lane Change**: When you engage a turn signal while Autosteer is active, Auto Lane Change moves Model Y into the adjacent lane in the direction indicated by the turn signal (see Auto Lane Change on page 116).
- **Navigate on Autosteer**: Navigate on Autosteer builds on the features of Traffic-Aware Cruise Control and Autosteer. While Autosteer is active, Navigate on Autosteer allows Model Y to suggest and, if configured, automatically change lanes to pass other vehicles and follow the navigation route (see Navigate on Autosteer on page 117).

  NOTE: Navigate on Autosteer is a BETA feature.

These features use information from the cameras on Model Y to detect lane markings, road edges, and other vehicles and road users around Model Y.

CAUTION: Ensure all cameras are clean and free of obstructions before each drive and before using the features described in this topic (see Cleaning a Camera on page 208). Dirty cameras and sensors (if equipped), as well as environmental conditions such as rain and faded lane markings, can affect performance. If a camera is obstructed or blinded, Model Y displays a message on the touchscreen and Self-Driving features may not be available.

CAUTION: It is your responsibility to familiarize yourself with the limitations of Autosteer and the situations in which driver intervention may be needed. For more information, see Self-Driving Limitations and Warnings on page 136.

Before Using Autosteer

Before you use Autosteer, customize how it works by touching **Controls** > **Self-Driving**.

- **Set Speed**: Choose whether Autosteer engages at the currently detected speed limit or your current driving speed. Touch **Controls** > **Self-Driving** and choose either **Speed Limit** or **Current Speed**.
- **Offset**: If you choose **Speed Limit**, you can specify an offset by touching **Set Speed Offset**. You can choose **Fixed** (the cruising speed adjusts by a specific amount on all roads) or **Percentage** (the cruising speed is adjusted as a percentage of the road's detected speed limit).
- **Green Traffic Light Chime**: In Canada and U.S.: If on, a chime will sound when you are waiting at a red traffic light and the light turns green. If you are not actively using Traffic-Aware Cruise Control and are waiting at a red light with a car in front of you, the chime sounds when the car ahead of you advances.

Autosteer

NOTE: *Depending on market region, vehicle configuration, options purchased, Self-Driving hardware, and software version, your vehicle may not be equipped with Autosteer, or the feature may not operate exactly as described.*

To enable Autosteer:

1. Touch **Controls** > **Self-Driving** > **Autosteer (Beta)**.
2. After carefully reading and understanding the popup window, touch **Yes**.

Autosteer is available when you are driving between 18 mph (30 km/h) and 85 mph (140 km/h). You can activate Autosteer at lower speeds if there is a vehicle detected at least 5 feet (1.5 meters) ahead of Model Y.

NOTE: To engage Autosteer, headlights must be set to **On** or **Auto**. Although Autosteer is available both during the day and in low light conditions (dusk or dark), it aborts or is unavailable if headlights are set to **Off**. When Autosteer is engaged, **Adaptive Headlights** (or, on some vehicles, **Auto High Beam**) is automatically enabled (see High Beam Headlights on page 81) and wipers are set to **Auto**.

To use Autosteer:

1. Press the right scroll button.

2. To change the set speed, roll the right scroll wheel up to increase, or down to decrease. For more information, see While Using Autosteer on page 119.
3. To cancel Autosteer, press the right scroll button or press the brake pedal. For more information, see Canceling Autosteer on page 118.

When you engage Autosteer, Model Y confirms activation with an audible chime and briefly displays a message on the touchscreen reminding you to pay attention to the road and be ready to take over at any time.

To indicate that Autosteer is available (but not actively steering Model Y), the top corner of the touchscreen displays a gray steering wheel icon next to the driving gear.

When a Autosteer is engaged and actively controlling Model Y, the steering wheel icon is blue.

When Autosteer is able to detect lane markings, it displays the edges of the driving lane in blue on the touchscreen.

Whenever Autosteer is active, Traffic-Aware Cruise Control is active as well.

In situations where the speed limit cannot be detected when Autosteer is engaged, Autosteer reduces your driving speed and limits the set cruising speed to 45 mph (70 km/h). Although you can manually accelerate to exceed the limited speed, Model Y will not brake for detected obstacles as long as you are applying the accelerator pedal. Autosteer slows down to the limited speed when you release the accelerator pedal. When you leave the road or disengage Autosteer by using the steering wheel, you can increase your set speed again, if desired.

WARNING: Autosteer is a hands-on assistance feature. Keep your hands on the steering wheel at all times, be mindful of road conditions and surrounding traffic, and always be prepared to take immediate action. Failure to follow these instructions could cause damage, serious injury or death. It is your responsibility to familiarize yourself with the limitations of Autosteer and the situations in which it may not work as expected. For more information, see Self-Driving Limitations and Warnings on page 136.

Auto Lane Change

NOTE: *Depending on market region, vehicle configuration, options purchased, Self-Driving hardware, and software version, your vehicle may not be equipped with Auto Lane Change, or the feature may not operate exactly as described.*

If you engage a turn signal while Autosteer is active, Model Y moves into the adjacent lane in the direction indicated by the turn signal, provided the following conditions are met:

- The turn signal is engaged.
- Lane markings indicate that a lane change is permitted.

- Midway through the lane change, Model Y must detect the target lane's outside lane marking. If this lane marking is not detected, the lane change is aborted and Model Y returns to its original driving lane.
- The view of the camera(s) is not obstructed.
- Model Y does not detect a vehicle in its blind spot, or a vehicle or obstacle up to the center of the target lane. If a vehicle or other obstacle is detected in the target lane, it is shown in red in the visualization on the touchscreen and Model Y does not complete the lane change until it is safe to do so.

WARNING: Although Auto Lane Change is designed to detect vehicles and obstacles in adjacent lanes, it is your responsibility to always perform visual checks to make sure it is safe and appropriate to move into the target lane. If Auto Lane Change cannot change lanes due to inadequate data, the touchscreen displays a series of warnings. Therefore, when using Auto Lane Change, always pay attention to the touchscreen and be prepared to manually steer Model Y.

The minimum speed at which Auto Lane Change changes lanes may vary depending on region, adjacent lane speeds, and other factors. Always be ready to manually steer and change lanes as necessary. When an automatic lane change is in progress, Overtake Acceleration is activated, allowing Model Y to accelerate closer to a vehicle in front (see Overtake Acceleration on page 121).

When you engage a turn signal, Auto Lane Change moves Model Y one lane at a time. Moving into an additional lane requires you to engage the turn signal a second time after the first lane change is complete.

As Model Y changes lanes, it is important to monitor its performance by watching the driving path in front of you and the surrounding area. Stay prepared to take over steering at any time. As you are crossing over into the adjacent lane, the touchscreen displays the location in the lane that Model Y is moving into.

Navigate on Autosteer

NOTE: *Depending on market region, vehicle configuration, options purchased, Self-Driving hardware, and software version, your vehicle may not be equipped with Navigate on Autosteer, or the feature may not operate exactly as described.*

To enable Navigate on Autosteer, touch **Controls** > **Self-Driving** > **Navigate on Autosteer (Beta)**. Then, to customize how you want Navigate on Autosteer to operate, touch **Customize Navigate on Autosteer**:

- **Enable at Start of Every Trip**: Choose whether to automatically enable Navigate on Autosteer for every navigation route. When enabled, the Navigate on Autosteer button on the turn-by-turn direction list is already enabled at the start of every trip.
- **Speed Based Lane Changes**: Navigate on Autosteer is designed to perform both route-based and speed-based lane changes. Speed-based lanes changes are optional. You can use this setting to disable speed-based lane changes or to specify how assertively you want Navigate on Autosteer to change lanes to achieve the set cruising speed (**Mild**, **Average**, or **Mad Max**).
- **Exit Passing Lane**: Choose whether you want Navigate on Autosteer to maneuver out of a passing lane when navigating to a destination. In addition to route-based and speed-based lane changes, Navigate on Autosteer requests a lane change out of a passing lane as a reminder to stay in a slower lane when you are not passing other vehicles. Choose **No** to disable this and keep Model Y in a passing lane except when needed to stay on the navigation route.
- **Require Lane Change Confirmation**: (if equipped) By default, Navigate on Autosteer requires your confirmation before proceeding with a lane change by engaging the appropriate turn signal. If you do not confirm the lane change within 3 seconds, a chime sounds to remind you that Navigate on Autosteer requires your confirmation to change lanes.
- **Lane Change Notification**: You can specify if or how you want to be notified of lane changes (**Off**, **Chime**, **Vibrate**, or **Both**).

If **Enable at Start of Every Trip** is turned on, Navigate on Autosteer engages automatically when:

- Autosteer is active.
- You are navigating to a destination.
- You are on a controlled-access highway.

Once enabled, the Navigate on Autosteer button appears on the map's turn-by-turn direction list whenever a navigation route is active and the route includes at least one controlled-access highway.

If **Enable at Start of Every Trip** is turned off, touch the **Navigate on Autosteer** button above the turn-by-turn directions to enable it. Once the Navigate on Autosteer is selected, it will engage whenever you engage Autosteer.

The Navigate on Autosteer icon shows in the turn-by-turn direction list when you are navigating to a destination and Navigate on Autosteer is available but not active.

If Navigate on Autosteer is active, the icon is blue. If **Enable at Start of Every Trip** is turned on, the Navigate on Autosteer icon is selected whenever you start navigation. Touch the icon to cancel Navigate on Autosteer and revert to Autosteer.

Whenever Navigate on Autosteer is active, the Navigate on Autosteer button is blue and the touchscreen displays the driving lane as a single blue line in front of Model Y:

The turn-by-turn directions display the Autosteer icon next to the maneuvers (such as off-ramps) that Navigate on Autosteer will handle.

When Navigate on Autosteer is engaged Model Y automatically makes both speed-based and route-based lane changes.

- **Speed Based Lane Changes:** Navigate on Autosteer changes lanes to reduce driving time to your destination. For example, if Model Y is behind a vehicle going below the set cruising speed, Navigate on Autosteer will move into the passing lane to pass it. Speed-based lanes changes are optional.
- **Route Based Lane Changes:** Navigate on Autosteer changes lanes to route you to your destination. For example, Navigate on Autosteer will move into the exit lane as Model Y approaches the off-ramp specified by the navigation route.

If you ignore a route-based lane change suggestion (for example, you are driving in the left lane while approaching an off-ramp on the right side of the highway), Navigate on Autosteer is unable to maneuver onto the off-ramp and as a result, you are re-routed to your destination.

NOTE: When determining navigation routes, and maneuvers at interchanges, Navigate on Autosteer considers whether or not you want to use High Occupancy Vehicle (HOV) lanes. Therefore, ensure the **Use HOV Lanes** setting is appropriate for your circumstances (see Maps and Navigation on page 169). If the setting is off, Navigate on Autosteer never uses a HOV lane, regardless of time of day. If the setting is on, Navigate on Autosteer uses HOV lanes, whenever applicable.

CAUTION: Navigate on Autosteer may not always attempt to exit at an off-ramp or change lanes, even when an exit or lane change is determined by the navigation route. Always remain alert and be prepared to manually steer onto an off-ramp, or make a lane change to prepare for, or to exit at, an off-ramp or interchange.

Navigate on Autosteer activates and deactivates based on the type of road you are driving on. When Navigate on Autosteer is active and you approach an off-ramp or interchange along your navigation route, the appropriate turn signal engages and Autosteer maneuvers Model Y onto the off-ramp or interchange.

When you leave a controlled-access highway Navigate on Autosteer reverts to Autosteer—a chime sounds and the touchscreen displays the driving lane lines in blue (instead of the single blue in front of Model Y). When Navigate on Autosteer deactivates, Autosteer remains active. Always be prepared to take appropriate action.

NOTE: Navigate on Autosteer is a hands-on feature. Keep your hands on the steering wheel at all times, be mindful of road conditions and surrounding traffic, and always be prepared to take immediate action. Failure to follow these instructions could cause damage, serious injury or death. It is your responsibility to familiarize yourself with the limitations of Navigate on Autosteer and the situations in which it may not work as expected. For more information, see Self-Driving Limitations and Warnings on page 136.

Canceling Autosteer

Autosteer cancels when:

- You press the right scroll button on the steering wheel.
- You press the brake pedal.
- You exceed 85 mph (140 km/h).
- You shift into Reverse, Park, or Neutral.
- A door is opened.
- An Automatic Emergency Braking event occurs (see Collision Avoidance Assist on page 143).
- The driver's seatbelt is released, and/or the driver gets out of their seat.

- You exceed 85 mph (140 km/h).
- You take over steering manually.
- You do not respond to repeated reminders to keep your hands on the wheel and subsequent messages on the touchscreen .

When Autosteer cancels, a chime sounds and the Autosteer icon either turns gray to indicate that Autosteer is no longer active, or disappears to indicate that it is not currently available.

Navigate on Autosteer cancels when Autosteer cancels, as described above. In addition, Navigate on Autosteer cancels when:

- You touch the Navigate on Autosteer button on the map's turn-by-turn direction list. In this case, Autosteer is still active.
- You leave a controlled-access highway. When this happens, Autosteer is still active.

When Navigate on Autosteer cancels but Autosteer remains active, a chime sounds and the visualization goes from a single blue line in the driving lane to two blue lines on either side of the lane.

When Autosteer cancels, Model Y does not coast. Instead, regenerative braking slows down Model Y in the same way as when you move your foot off the accelerator when driving normally (see Regenerative Braking on page 86).

While Using Autosteer

When Autosteer is active and maintaining a set speed, the speed is highlighted with blue text on the touchscreen.

When Autosteer is active, the steering wheel icon is blue and the lane markings are highlighted in blue on the visualization. If Navigate on Autosteer is also active, the Navigate on Autosteer button is blue and the touchscreen displays the driving lane as a single blue line in front of Model Y.

If unable to detect lane markings, Autosteer may determine the driving lane based on a vehicle you are following. In most cases, Autosteer attempts to center Model Y in the driving lane. However, there may be situations in which Autosteer follows a driving path that is offset from the center of the lane (for example, if guard rails are detected).

Maintaining the Set Speed

When Autosteer is active, Model Y maintains your set cruising speed whenever a vehicle is not detected in front of it. When cruising behind a vehicle, Model Y accelerates and decelerates as needed to maintain a chosen following distance (see Adjusting the Following Distance on page 119), up to the set speed.

You can manually accelerate at any time by pressing the accelerator pedal, but when you release the pedal Model Y resumes cruising at the set speed.

Model Y also adjusts the cruising speed when entering and exiting curves.

When Model Y is actively slowing down to maintain the selected distance from the vehicle ahead, brake lights turn on. You may notice slight movement of the brake pedal. However, when Model Y is accelerating, the accelerator pedal does not move.

Changing the Set Speed

Roll the right scroll wheel up to increase, or down to decrease, the set speed.

To change the cruising speed to the current speed limit (including any offset you've specified), press the speed limit sign on the touchscreen.

It may take a few seconds for Model Y to reach the new cruising speed.

NOTE: On a residential road, a road without a center divider, or a road that is not controlled access, the maximum allowed cruising speed is limited and the touchscreen displays a message. The restricted speed will be the speed limit of the road plus 5 mph (10 km/h).

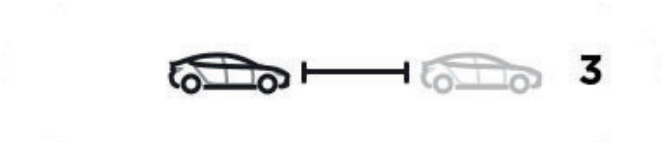
Adjusting the Following Distance

To adjust the following distance you want to maintain between Model Y and a vehicle traveling ahead of you, press the steering wheel's right scroll button to the left or right.

The closest following distance is 2.

Each setting corresponds to a time-based distance that represents how long it takes for Model Y, from its current location, to reach the location of the rear bumper of the vehicle ahead of you. Autosteer retains your setting until you change it again.

As you adjust the following distance, the touchscreen displays the current setting.



Stopping and Slowdowns

When moving significantly faster than vehicles in adjacent lanes, Model Y automatically reduces the driving speed. This is especially helpful in heavy traffic situations or when vehicles are constantly merging into different lanes. When Model Y detects other vehicles driving significantly slower, the touchscreen highlights the adjacent lanes with arrows and detected vehicles in gray, and Model Y reduces the driving speed as appropriate. To temporarily override this feature, press the accelerator pedal.

When following a vehicle, Autosteer remains active at low speeds, even when Model Y comes to a full stop. For example, Autosteer remains active even if Model Y slows down to a complete or near-complete stop in heavy, stop-and-go traffic on a highway. When traffic starts moving more rapidly, Autosteer again accelerates up to the set speed.

Sometimes when Model Y is at a full stop, Autosteer goes into a HOLD state. If this happens, briefly press the accelerator pedal to resume cruising.

When the HOLD status is active, the touchscreen displays the HOLD icon and a message that indicates that you need to resume cruise control.

Model Y goes into HOLD state while Autosteer is active in the following circumstances:

- Model Y has been at a standstill for 5 minutes.
- Model Y detects a pedestrian (the HOLD state may clear when the pedestrian is no longer detected).
- Model Y suddenly loses visibility of the vehicle in front of you.
- An obstacle is detected in front of Model Y.

Cruising Near or On Exits

When you are cruising near an exit on a controlled-access highway and engage the turn signal toward the off-ramp, Autosteer assumes you are exiting and begins to slow down Model Y. If you do not drive onto the off-ramp, Autosteer resumes cruising at the set speed.

In a region with right hand traffic, this occurs only when you engage the right turn signal when driving in the right-most lane within 164 ft. (50 meters) of an exit. Likewise in regions with left hand traffic, this occurs when engaging the left turn signal when driving in the left-most lane within 164 ft. (50 meters) of an exit.

NOTE: If Navigate on Autosteer is active, Model Y will perform a route-based lane change to enter the exit lane and take the off-ramp as necessary to follow the navigation route.

When enabled while on a highway interchange or off-ramp, Traffic-Aware Cruise Control may reduce your set speed in 5 mph (5 km/h) increments – to as slow as 25 mph (40 km/h) – to better match the reported speeds of other Tesla vehicles that have driven at that specific location. To override this and continue cruising at your set speed, tap the accelerator pedal. The new set speed is maintained for the duration of the interchange or off-ramp (unless you override it or cancel Autosteer). After the interchange or off-ramp, the set speed may revert or change as necessary based on the new location. For example, if you merged onto a different highway, the set cruising speed reverts to what it was before driving on the interchange.

WARNING: In some cases (such as having insufficient data), Traffic-Aware Cruise Control may not automatically reduce the set speed on the highway interchange or off-ramp. Do not rely on Traffic-Aware Cruise Control to determine an appropriate driving speed. Tesla recommends driving at a speed that is safe for road conditions and within posted speed limits.

When cruising onto an on-ramp to a controlled-access highway, Autosteer automatically adjusts the set cruising speed to the speed limit of the highway, plus any offset you have specified. If Navigate on Autosteer is engaged, it disengages as you leave the controlled-access highway (see Canceling Autosteer on page 118). In this case, Autosteer remains active.

Overtake Acceleration

Engage the turn signal momentarily to accelerate Model Y towards the vehicle ahead of it. By momentarily holding the turn signal, you can quickly accelerate up to your set speed without having to press the accelerator pedal as long as:

- Traffic-Aware Cruise Control is operating and detects a vehicle in front of you.
- No obstacles or vehicles are detected in the target lane.
- Model Y is traveling below the set speed, but over 45 mph (72 km/h).

NOTE: If Autosteer is active and you fully engage the turn signal, Model Y will change lanes automatically (see Auto Lane Change on page 116).

Model Y stops accelerating when you reach your set cruising speed, if changing lanes takes too long, or if Model Y gets too close the vehicle ahead. Model Y also stops accelerating if you disengage the turn signal.

Stop Light and Stop Sign Warning

While Autosteer is in use, Model Y displays a warning on the touchscreen and sounds a chime if it detects that you are likely to run through a red stop light or stop sign. If this happens, **TAKE IMMEDIATE CORRECTIVE ACTION!**

The visual and audible warnings cancel after a few seconds or when you press the brake pedal, whichever comes first.

Stop Light and Stop Sign Warning provides warnings only. It does not slow down or stop Model Y at red traffic lights, stop signs, road markings, etc. If equipped with Traffic Light and Stop Sign Control, you can enable this feature to automatically stop Model Y at traffic lights and stop signs (see Traffic Light and Stop Sign Control on page 123).

Emergency Vehicles

If available in your market region, Model Y automatically reduces driving speed when lights from an emergency vehicle are detected when using Autosteer at night on a high speed road. When this happens, the touchscreen displays a message informing you of the slowdown. You will also hear a chime, and see a reminder to keep your hands on the steering wheel. When the light detections pass by or cease to appear, Autosteer resumes your cruising speed. Alternatively, you may tap the accelerator to resume your cruising speed.

Never depend on Autosteer features to determine the presence of emergency vehicles. Model Y may not detect lights from emergency vehicles. Keep your eyes on your driving path and always be prepared to take immediate action.

Take Over Immediately

In situations where Self-Driving is unable to steer Model Y, a warning chime sounds and the touchscreen displays the following message.

Take over immediately

When you see this message, **take over steering immediately**.

Driver Attentiveness

Autosteer requires driver attentiveness. **Your hands must be on the steering wheel at all times while Autosteer is engaged, and you must monitor your surroundings, the road, and other road users.**

Autosteer periodically displays a message reminding the driver to apply slight force to the steering wheel:

Apply slight turning force to steering wheel

Autosteer detects your hands by recognizing slight resistance as the steering wheel turns, or from you manually turning the steering wheel very lightly (without enough force to take over steering). Autosteer also qualifies your hands as being detected if you engage a turn signal use the Autosteer stalk, or use a button or scroll wheel on the steering wheel. If Model Y does not detect slight turning force on the steering wheel, the touchscreen flashes and eventually chimes repeatedly. Do not use handheld devices while Autosteer is engaged.

If you repeatedly ignore prompts to apply slight force to the steering wheel or to pay attention, Autosteer disables for the rest of the drive and displays the following message requesting you to drive manually. If you don't resume manual steering, Autosteer sounds a continuous chime, turns on the warning flashers, and slows the vehicle to a complete stop.

Self-Driving temporarily disabled
Driver attention warnings ignored

Self-Driving Suspension

Use of Self-Driving features will be suspended if improper usage is detected.

Use of Autosteer (if equipped) and Full Self-Driving (Supervised) is suspended for a week when you or another driver of your vehicle receives five "strikeouts." A strikeout is when the Self-Driving system disengages for the remainder of a trip after the driver receives several audio and visual warnings for inattentiveness.

You can see how many strikeouts are remaining before Self-Driving access is suspended by touching **Controls** > **Self-Driving**.

A strikeout is forgiven after half a week (3.5 days), as long as you don't receive another strikeout in that time.

NOTE: If your access to Autosteer (if equipped) and Full Self-Driving (Supervised) is suspended, you also lose access to Autopark. However, you can still use Traffic-Aware Cruise Control and all active safety features are still enabled.

There may be occasions where driver intervention is required and you must take over immediately to maintain safe driving. Driver-initiated disengagements do not count as improper usage and are expected from the driver.

Traffic Light and Stop Sign Control

NOTE: *Depending on market region, vehicle configuration, options purchased, Self-Driving hardware, and software version, your vehicle may not be equipped with Traffic Light and Stop Sign Control, or the feature may not operate exactly as described. Traffic Light and Stop Sign Control is available for select Tesla vehicles. To check if your vehicle is equipped with a specific Self-Driving feature, refer to the Owner's Manual on your vehicle's touchscreen or touch* ***Controls*** *>* ***Self-Driving****. Feature availability depends on vehicle configuration, region, model, vehicle trim and model year.*

NOTE: ***Traffic Light and Stop Sign Control is a BETA feature and works best on roads that are frequently driven by Tesla vehicles. Traffic Light and Stop Sign Control attempts to stop at all traffic lights and may also stop at green lights.***

Traffic Light and Stop Sign Control is designed to recognize and respond to traffic lights and stop signs, slowing Model Y to a stop when using Traffic-Aware cruise control or Autosteer. This feature uses the vehicle's forward-facing cameras, in addition to GPS data, and slows the car for all detected traffic lights, including green, blinking yellow, and off lights in addition to stop signs and some road markings. As Model Y approaches an intersection, the touchscreen displays a notification indicating the intention to slow down. You must confirm that you want to continue or Model Y stops at the red line displayed on the touchscreen's driving visualization.

WARNING: NEVER make assumptions and predict when and where Traffic Light and Stop Sign Control will stop or continue through an intersection or road marking. From a driver's perspective, the behavior of Traffic Light and Stop Sign Control may appear inconsistent. Always pay attention to the roadway and be prepared to take immediate action. It is the driver's responsibility to determine whether to stop or continue through an intersection. Never depend on Traffic Light and Stop Sign Control to determine when it is safe and/or appropriate to stop or continue through an intersection.

Before Using

Before using Traffic Light and Stop Sign Control, you must:

- Ensure that forward-facing cameras are unobstructed (see) and calibrated (see Drive to Calibrate Cameras on page 20). Traffic Light and Stop Sign Control depends on the ability of the cameras to detect traffic lights, stop signs, and road markings.
- Ensure that the latest version of maps has been downloaded to Model Y. Although Traffic Light and Stop Sign Control primarily uses visual data received from the vehicle's cameras, greater accuracy is achieved when using the most recent map data. To check which version of maps is currently downloaded, touch **Controls** > **Software**. You must connect to a Wi-Fi network to receive updated maps (see Map Updates on page 175).
- Enable the feature. With the vehicle in Park, touch **Controls** > **Self-Driving** and then touch **Traffic Light and Stop Sign Control**. Once enabled, Traffic Light and Stop Sign Control operates whenever Traffic-Aware Cruise Control or Autosteer is active.

How it Works

When Traffic Light and Stop Sign Control is enabled and you are using Traffic-Aware Cruise Control, Autosteer, Full Self-Driving (Supervised), the touchscreen displays a popup message to inform you that an upcoming traffic light, stop sign, or road marking has been detected. As it approaches the stop location, **even at an intersection where the traffic light is green,** Model Y slows down and displays a red line to indicate where Model Y will stop. To continue through the intersection—even if the traffic light is green —you must briefly press the accelerator pedal to give the vehicle permission to proceed. When you've confirmed that you want to proceed, the red stop line turns gray and Model Y continues through the intersection and resumes your set cruising speed.

NOTE: If Model Y is approaching a green light and detects that a vehicle in front of you is continuing through the intersection, Model Y continues through the intersection without requiring your confirmation, provided you are not in a turning lane and the vehicle can detect that your hands are on the steering wheel.

NOTE: If, after you briefly press the accelerator pedal to confirm that you want to continue through the intersection, the traffic signal changes before you enter the intersection (for example, the light changes from green to yellow or from yellow to red), Model Y may determine that it is not appropriate to proceed. Therefore, Model Y stops and you must press the accelerator to proceed. At all times, it is your responsibility to ensure the vehicle stops or accelerates appropriately and safely.

WARNING: Traffic Light and Stop Sign Control DOES NOT turn Model Y through an intersection unless Full Self-Driving (Supervised) is engaged. When in a turning lane, Model Y stops at the red stop line. To proceed, briefly press the accelerator pedal—Model Y continues *straight* through the intersection (even when in a turning lane), so you MUST manually steer Model Y through the intersection (which cancels Autosteer).

Traffic Light and Stop Sign Control is designed to operate as described only when the following conditions are met:

- One of the following Self-Driving features is engaged:
  - Traffic-Aware Cruise Control
  - Autosteer
  - Full Self-Driving (Supervised)
- The cameras can detect an upcoming traffic light, stop sign or road marking (for example, cameras are unobstructed and have a clear line-of-sight to the traffic light, stop sign, or road marking).

- The touchscreen on Model Y is displaying an upcoming traffic light in "bold" format. Model Y does not acknowledge traffic lights that the touchscreen shows as faded. If a traffic light is not directly ahead of the camera (for example, it is located at an angle of the camera's view, or located in an adjacent lane) the touchscreen displays it as faded and Model Y does not slow down and stop for it.

WARNING: If the touchscreen is not displaying a red stop line at an upcoming intersection, Model Y does not slow down or stop. It is the driver's responsibility to pay attention to upcoming intersections and monitor traffic conditions to determine when and if the vehicle should stop and then to take appropriate action as needed.

WARNING: Never depend on Traffic Light and Stop Sign Control to determine whether to stop at, or proceed through, an intersection. Drive attentively by watching the road and paying attention to the roadway, upcoming intersections, traffic conditions, crosswalks, and other road users. It is always the driver's responsibility to determine whether to stop or proceed. Be prepared to take immediate action. Failure to do so can result in injury or death.

WARNING: In some situations, Traffic Light and Stop Sign Control may inaccurately detect a traffic light or stop sign, causing Model Y to slow down unexpectedly. Be prepared to take immediate action at all times.

WARNING: You must briefly press the accelerator pedal to confirm that you want to proceed through an intersection, regardless of the status of the traffic light. If you do not confirm, Model Y stops at the red stop line displayed on the touchscreen, even if stopping may be inappropriate. Stopping at a green light may confuse other drivers and may result in a collision, injury or death. Therefore, always pay attention to upcoming intersections and be prepared to manually brake or accelerate in response to surroundings.

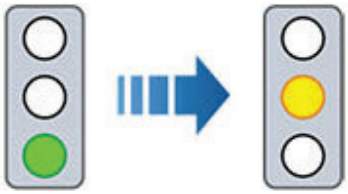
WARNING: Never assume that your ability to see a traffic light, stop sign, or road marking (especially at a complex intersection, or an intersection in which a traffic light or sign is partially obstructed, etc.) means that Model Y can also see it and respond appropriately.

WARNING: Even the most recent map data does not include all traffic lights and stop signs. Therefore, Traffic Light and Stop Sign Control relies heavily on the ability of the cameras to detect traffic lights, stop signs, road markings, etc. As a result, Model Y may ignore an intersection that is blocked from the camera's view (for example, obstructed by a tree or a large vehicle or object, or located near a steep hill or sharp curve).

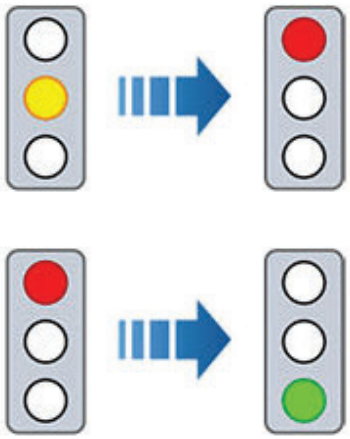
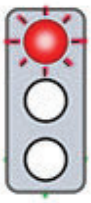
WARNING: Traffic Light and Stop Sign Control is not a substitute for attentive driving and sound judgment.

Traffic Lights

When driving with Autosteer or Traffic-Aware Cruise Control engaged, and Traffic Light and Stop Sign Control enabled, Model Y is designed to respond as follows when approaching intersections controlled by a traffic light:

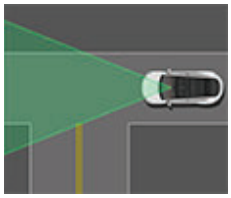
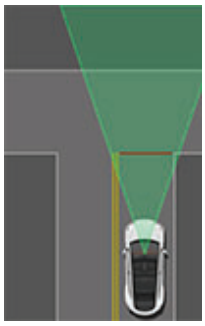
Type of Traffic Light	Vehicle Intended Response
	At a solid green traffic light, or at a traffic light that is currently off (not illuminated), Model Y slows down. If you are following a car in front of you that continues through the intersection, the touchscreen displays a green stop line and provided your hands are detected on the steering wheel, Model Y also continues. If a car is not in front of you, the touchscreen displays a red stop line. You must confirm that you want to continue through the intersection by briefly pressing the accelerator pedal. If you don't confirm, Model Y stops at the red stop line displayed on the touchscreen. **NOTE:** Model Y resumes the set cruising speed when it continues through the intersection, taking into consideration the speed of a vehicle in front of you.
	Model Y slows down and comes to a complete stop at the red stop line displayed on the touchscreen. When you want to continue through the intersection (for example, the light turns green again, or once Model Y has come to a complete stop), you must briefly press the accelerator pedal.
	Model Y slows down and comes to a complete stop at the red stop line displayed on the touchscreen. When you want to proceed through the intersection (for example, the light turns green again), you must briefly press the accelerator pedal. **NOTE:** If the traffic light changes *after* you've confirmed that you want to proceed (for example, a green traffic light turns yellow), Model Y may stop instead of continuing, especially if Model Y determines that it can safely stop before entering the intersection. **NOTE:** Model Y is not designed to proceed through an intersection when the traffic light is red or if the light turns yellow in situations when there is adequate distance to safely stop before entering the intersection. **NOTE:** You can take over driving at any time by manually braking to cancel Autosteer or Traffic-Aware Cruise Control.

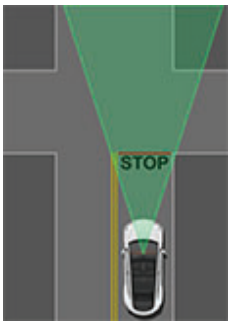
Traffic Light and Stop Sign Control

Type of Traffic Light	Vehicle Intended Response
	
	Model Y slows down. To proceed, you must briefly press the accelerator pedal. If you don't, Model Y stops at the red stop line displayed on the touchscreen. **NOTE:** To prevent Model Y from stopping, and to minimize how much it slows down as it approaches, you can confirm that you want to proceed by briefly pressing the accelerator pedal at any time after the touchscreen displays the red stop line. Model Y resumes your set cruising speed immediately after you confirm (taking into consideration the speed of a vehicle in front of you). **WARNING:** Approach attentively and be prepared to press the brake pedal to slow down or stop.
	Model Y slows down and comes to a complete stop at the red stop line displayed on the touchscreen. When you want to proceed through the intersection (for example, traffic laws and conditions indicate it is safe and legal to proceed), you must briefly press the accelerator pedal.

Stop Signs and Road Markings

When driving with Autosteer or Traffic-Aware Cruise Control engaged, and Traffic Light and Stop Sign Control enabled, Model Y is designed to respond as follows when approaching intersections controlled by stop signs, stop lines, or road markings:

Type of Intersection	Vehicle Intended Response
 No Traffic Control  Arm of T-junction	Model Y assumes the right of way and continues straight without slowing down or stopping.
 End of T-junction	If Model Y detects a T-junction based on the map data, Model Y slows down and comes to a complete stop at the red stop line displayed on the touchscreen. When you want to proceed, you must take over steering and acceleration. **WARNING:** Model Y may not stop at a T-junction that does not have a stop sign or stop line, or if the T-junction is not included in the map data. Drive attentively and be prepared to stop (when necessary and/or appropriate).
  Stop Sign	Model Y slows down and comes to a complete stop at the red stop line displayed on the touchscreen. When you want to proceed through the intersection, you must briefly press the accelerator pedal. **NOTE:** If you confirm that you want to proceed through an intersection controlled by a stop sign by briefly pressing the accelerator pedal before Model Y has stopped, your confirmation is ignored. Model Y is not designed to proceed through a stop sign without stopping. **NOTE:** Even when using Autosteer, and even if you have engaged a turn signal, you must turn the steering wheel yourself (which cancels Autosteer) to complete a turn at an intersection.

Type of Intersection	Vehicle Intended Response
Stop Sign and Road Marking	
 Road Marking	

WARNING: Model Y also slows down and stops at a roundabout. You must take over steering (which cancels Autosteer) and briefly press the accelerator pedal to confirm that you want to continue through the roundabout.

WARNING: At crosswalks, Model Y may slow down and may stop, depending on whether the crosswalk is controlled by a traffic light and whether the cameras detect pedestrians, bicyclists, etc. in the crosswalk. Pay particular attention at crosswalks and be prepared to take over at any time. Failure to do so can result in injury or death.

Limitations

Depending on many different circumstances and environmental conditions, Traffic Light and Stop Sign Control *may or may not* stop at:

- Railroad crossings.
- Keep-out zones.
- Toll booths.
- Crosswalk systems.
- Yield signs or temporary traffic lights and stop signs (such as at construction areas).
- Miscellaneous traffic U-turn lights, bicycle and pedestrian crossing lights, lane availability lights, etc.

In addition, Traffic Light and Stop Sign Control is particularly unlikely to operate as intended, can disengage, or may not operate, when one or more of the following conditions are present:

- Driving through consecutive light-controlled intersections that are very close to each other.
- Visibility is poor (heavy rain, snow, fog, etc.) or weather conditions are interfering with camera or sensor operation.
- Bright light (such as direct sunlight) is interfering with the view of the camera(s).
- A camera is obstructed, covered, damaged, or not properly calibrated.

- Driving on a hill or on a road that has sharp curves on which the cameras are unable to see upcoming traffic lights or stop signs.
- A traffic light, stop sign, or road marking is obstructed (for example, a tree, a large vehicle, etc.).
- Model Y is being driven very close to a vehicle in front of it, which is blocking the view of a camera.

WARNING: The limitations listed above are not an exhaustive list of reasons why Model Y may not operate as expected. Many unforeseen circumstances can adversely impact the accurate operation of Traffic Light and Stop Sign Control. Using this feature does not reduce or eliminate the need to drive attentively and responsibly. You must be prepared to take appropriate and immediate action at all times.

Autopark

NOTE: *Depending on market region, vehicle configuration, options purchased, Self-Driving hardware, and software version, your vehicle may not be equipped with Autopark.*

Autopark uses data to simplify parking on public roads by maneuvering Model Y into parallel and perpendicular parking spaces.

NOTE: Autopark is unavailable when Traffic-Aware Cruise Control is enabled. To use Autopark, enable Autosteer or Full Self-Driving (Supervised) by touching **Controls** > **Self-Driving**.

If your access to Autosteer and Full Self-Driving (Supervised) is suspended, you also lose access to Autopark. For more information, see Self-Driving Suspension on page 122.

WARNING: It is your responsibility to familiarize yourself with the limitations of Autopark and the situations in which it may not work as expected. For more information, see Self-Driving Limitations and Warnings on page 136.

CAUTION: Do not use Autopark if anything, such as a ball hitch, bike rack, or trailer, is attached to the tow hitch. Autopark may not stop for hitches when parking between or in front of other vehicles.

CAUTION: Autopark's performance depends on the ability of the cameras to determine the vehicle's proximity to curbs, objects, and other vehicles. Ensure all cameras are clean and free of obstructions before each drive and before using the features described in this topic (see Cleaning a Camera on page 208). Dirty cameras and sensors (if equipped), as well as environmental conditions such as rain and faded lane markings, can affect performance. If a camera is obstructed or blinded, Model Y displays a message on the touchscreen and Self-Driving features may not be available.

Parameters

Autopark detects potential parking spaces based on the following parameters:

Perpendicular Parking

- Your driving speed must be below 8 mph (13 km/h). If driving too fast, Autopark may not be able to accurately detect your desired parking space.
- The parking space must be at least as wide as your vehicle.
- The parking space must have at least three visible lines for the vehicle to park into, such as parking lines, road markings, or distinct curbs. Autopark may not work in a garage, for example, without three visible parking lines.
- Autopark may not work with textured road surfaces such as cobblestone or brick.

Parallel Parking

- Your driving speed must be below 8 mph (13 km/h). If driving too fast, Autopark may not be able to accurately detect your desired parking space.
- There must be a vehicle in front of or behind the space you want to park in.

NOTE: Autopark does not operate on angled parking spaces.

To Use Autopark

When driving, follow these steps to allow Autopark to maneuver Model Y into a parking space:

1. While driving slowly, monitor the touchscreen (after making sure that it is safe to do so) to see potential parking spaces detected by Self-Driving.

   NOTE: The detected parking spaces appear only if the vehicle's position and/or the circumstances of the surrounding area are such that Autopark can determine an appropriate driving path. If Autopark cannot determine an appropriate path (for example, when driving on a narrow street where moving into the parking space causes the front of the vehicle to extend into the adjacent lane), you can either reposition the vehicle, find a different parking space, or park manually.

2. Choose a spot, then release the steering wheel and touch **Start** on the touchscreen.
3. Autopark displays a message when parking is complete.

If you press the brake pedal when Autopark is actively parking Model Y, the parking process cancels.

CAUTION: Model Y does not brake if you choose to override the current speed by pressing the accelerator pedal while Autopark is active. In this case, Autopark cancels if you exceed 6 mph (10 km/h).

WARNING: Never depend on Autopark to find a parking space that is legal, suitable, and safe. Autopark may not always detect objects in the parking space. Always perform visual checks to confirm that a parking space is appropriate and safe.

WARNING: When Autopark is actively steering Model Y:

- Do not interfere with the movement of the steering wheel. Doing so cancels Autopark.
- Continually check your surroundings. Be prepared to apply the brakes to avoid vehicles, pedestrians, or objects.
- Monitor the touchscreen to ensure that you are aware of the instructions that Autopark is providing.

To Cancel Parking

Autopark cancels the parking sequence when:

- You manually move the steering wheel.
- You shift.
- You press the brake pedal.
- You press the right scroll button on the steering wheel.
- The parking sequence exceeds seven moves.

Take Over Immediately

In situations where Autopark is unable to steer Model Y, Autopark sounds a warning chime and displays the message **Take Over Immediately**.

This happens when:

- Model Y detects that a door is opened or that the driver is exiting the vehicle.
- One or more of the cameras is damaged, dirty, obstructed (such as by mud, ice, or snow, or by adhesive products such as wraps, stickers, etc.), or has poor visibility (due to heavy rain, snow, hail, etc. or poor lighting).
- Your speed exceeds 6 mph (10 km/h).
- An Automatic Emergency Braking event occurs (see Collision Avoidance Assist on page 143).

When you see this message, **take over immediately**.

Summon

NOTE: *Depending on market region, vehicle configuration, options purchased, Self-Driving hardware, and software version, your vehicle may not be equipped with Actually Smart Summon and Dumb Summon, or the features may not operate exactly as described.*

Summon allows you to park and retrieve Model Y using the Tesla mobile app while you are standing outside of your vehicle.

Summon includes both Dumb Summon and Actually Smart Summon (ASS).

Dumb Summon moves Model Y straight forward and reverse into, or out of, a parking spot.

Actually Smart Summon is designed to move Model Y to your location (using your phone's GPS as a target destination) or to a location of your choice, maneuvering around and stopping for objects as necessary.

While Summon is active, it is recommended that you maintain clear line of sight between you and Model Y. Closely monitor the vehicle and its surroundings at all times and familiarize yourself with the Limitations and Warnings on page 134.

Before Using Summon

1. Enable Dumb Summon and Actually Smart Summon by touching **Controls** > **Self-Driving** > **ASS (Actually Smart Summon)**. After carefully reading and understanding the pop-up window, touch **Yes**.
2. (Optional) Customize Summon settings.
   - **Completion Sound**: Select a sound to be played externally upon Model Y reaching its destination. Requires a Pedestrian Warning System (see Pedestrian Warning System on page 98).
   - **Emissions on Abort**: When enabled, Model Y performs an emissions system test if your Summon session is aborted. Requires a Pedestrian Warning System (see Pedestrian Warning System on page 98).
   - **Bumper Clearance** (Dumb Summon only): Set the distance that you want Dumb Summon to stop from a detected object. Note that this distance applies only to objects that Dumb Summon detects directly in front of Model Y when moving forward, or directly behind Model Y when reversing.
   - **Require Continuous Press** (Dumb Summon only): Requires that you press and hold a button on the mobile app to continue moving Model Y. Model Y stops when the button is released.

     NOTE: Require Continuous Press is always enabled.
3. (Optional) Enable **Standby Mode**: To keep Model Y ready to summon and reduce the time it takes to warm up, turn on Standby Mode. Additional battery power may be consumed while Standby Mode is active. When **Standby Mode** is turned on, you can conserve Battery energy by disabling Standby Mode at these locations (see Home, Work, and Favorite Destinations on page 172):
   - **Exclude Home** - Disables Standby Mode at the location you set as Home in your Favorites list.
   - **Exclude Work** - Disables Standby Mode at the location you set as Work in your Favorites list.
   - **Exclude Favorites** - Disables Standby Mode at any location in your Favorites list.

   To conserve energy, Summon automatically exits **Standby Mode** from midnight to 6:00 am and when Model Y is in Low Power Mode (see Low Power Mode on page 195). During these hours, a delay occurs as Summon starts up.

   If Model Y remains in Park for over 24 hours, **Standby Mode** disables until the vehicle is next driven.
4. Download the latest version of the Tesla mobile app. *Summon requires Tesla mobile app version 4.38.0 or later.*
5. Ensure that:
   - Your phone is connected to Model Y and that you are located within approximately 279 ft. (85 meters) (see Operating Range on page 132).
   - You are on private property. Summon is designed and intended for use only on parking lots and driveways located on private property where the surrounding area is familiar and predictable.
   - The vehicle's cameras are fully calibrated (see Drive to Calibrate Cameras on page 20).
   - You have a clear line of sight to Model Y.
   - Model Y is in Park, not charging, and all doors and trunks are closed.
   - All four tire pressures are 34 PSI or higher, as reported by the Tire Pressure Monitoring System (TPMS). For more information about checking and maintaining tire pressures, see Tire Care and Maintenance on page 201.

Operating Range

To use Summon, your phone must be connected to Model Y and within 279 ft. (85 meters) of the vehicle.

If Summon moves Model Y more than 279 ft. (85 meters) away from the connected phone while Summon is active, vehicle speed is limited to 3 mph (5 km/h).

If Model Y is more than 328 ft. (100 meters) away from the connected phone, Summon pauses, and if Model Y is over 345 ft. (105 meters) away, Summon comes to a stop and completes.

In any of the cases above, the Tesla mobile app provides haptic feedback and displays a message informing you that Model Y is close to, or has exceeded, the operating range of the feature. If moving the vehicle to a destination away from you, you may need to follow the car to stay within the operating distance.

NOTE: If Summon is operating for more than 7.5 minutes continuously, Summon aborts and Model Y shifts into Park.

Using Summon

Follow these steps to use Summon to park or retrieve your Model Y.

The following steps describe how to operate both Dumb Summon, which moves Model Y straight into or out of a parking space, and Actually Smart Summon, which allows Model Y to navigate around obstacles to come to you or to location of your choosing.

1. (Dumb Summon only) Align Model Y with the parking space so that Model Y can follow a straight path into or out of the space in either Drive or Reverse.
2. Open the Tesla mobile app and press **Summon**.

   The mobile app displays live camera feeds from Model Y to assist you in determining if the vehicle's surroundings are clear.

   NOTE: In order to view live camera feeds, your mobile device must be set up as a phone key to Model Y (see Keys on page 22). Whenever you are viewing the camera feeds in the Tesla mobile app, the headlights flash periodically. If your vehicle is equipped with Premium Connectivity, the live camera feed shares the same time limit as the **View Live Camera** feature of Sentry Mode, which is 1 hour (or 15 minutes for some regions) of cumulative usage per day (see Sentry Mode on page 154). If your vehicle is not equipped with Premium Connectivity, the camera stream is limited to 2 minutes. However, there is no limit to the camera stream when Summon is actively moving Model Y, regardless of whether or not Model Y is equipped with Premium Connectivity.

3. (Actually Smart Summon only) **Come to Me**: Position yourself anywhere within the blue circle where you have a clear line of sight to Model Y.

   Touch to select **Come to Me** mode. When selected, the icon is blue.

4. (Actually Smart Summon only) **Go to Target**: Touch the crosshair icon, then drag the map to position the pin on a chosen destination. To subsequently change the location, lift your finger, reposition the map, then press and hold **Go to Target** again.

   Touch to select **Go to Target** mode. When selected, the icon is blue.

5. Initiate Summon by pressing and holding:
   - **Forward** or **Reverse** (Dumb Summon).
   - **Come to Me** (Actually Smart Summon).
   - **Go to Target** (Actually Smart Summon).
6. Model Y shifts into Drive or Reverse, then moves slowly straight forward or in reverse (if you are using Dumb Summon), to your location (**Come to Me**), or to the selected location (**Go to Target**). In addition, headlights (as well as dome lights and, if equipped, fog lights) are set to **On** and wipers are set to **Auto**.

   If you are using **Come to Me** or **Go to Target**, Model Y navigates obstacles as needed. If you are using Dumb Summon, Model Y attempts to avoid obstacles while staying very close to its original path. Dumb Summon is not designed to steer around obstacles significantly blocking its path. If Model Y is unable to steer around an obstacle, Dumb Summon comes to a stop and completes.

   The mobile app will show you some of the camera feeds from Model Y for better supervision. Still, it is recommended to have a direct line of sight to the vehicle while summoning.

7. To stop Model Y at any time, simply release the **Forward** or **Reverse** button, or the **Come to Me** or **Go to Target** button.

   To resume Dumb Summon, once again hold down the **Forward** or **Reverse** button in the Tesla mobile app.

   Likewise, to resume an Actually Smart Summon session, simply press the **Come to Me** or **Go to Target** button again.

8. (Dumb Summon only) When Model Y is in the desired position, release the **Forward** or **Reverse** button. When you release the button, Model Y stops moving immediately and will shift into Park after several seconds.
9. (Actually Smart Summon only) When Model Y has arrived at your location (in **Come to Me** mode) or to the chosen location (in **Go to Target** mode), the hazard warning lights flash for several seconds, Model Y shifts into Park, and the mobile app displays a message indicating that Summon has completed.

   WARNING: When you release the button to stop Model Y, a slight delay occurs before the vehicle stops. Therefore, it is critical that you pay close attention to the vehicle's driving path at all times and proactively anticipate obstacles that the vehicle may be unable to detect.

Summon cancels, and requires you to restart it, when:

- A door handle is engaged or a door is opened.
- You interact with the steering wheel, brake pedal, accelerator pedal, or shift.
- The driving path is blocked.
- Model Y has moved the maximum distance of 1,558 ft. (475 meters), or has been moving for more than 7.5 minutes, since the start of the Summon session.

NOTE: If Summon moves Model Y forward 6 ft. (2 meters) and then backwards 3 ft. (1 meters), this is considered 9 ft. (3 meters) of travel.

- Your phone enters sleep mode or loses connectivity to Model Y.
- Model Y detects a safety risk.

Limitations and Warnings

Summon is disabled when:

- Model Y is in Valet Mode (see Valet Mode on page 95).
- Model Y is in Dog Mode (see Keep Climate On, Dog, and Camp on page 160).
- Model Y is in Drag Strip Mode (if equipped).

In addition, Actually Smart Summon is disabled on public roads. Actually Smart Summon is designed and intended for use only on parking lots and driveways located on private property where the surrounding area is familiar and predictable.

WARNING: It is still possible to use Dumb Summon on public roads. Use Dumb Summon on public roads only with extreme caution. Do not use Dumb Summon to enter the roadway.

Summon's performance depends on the ability of the cameras to determine the vehicle's proximity to objects, people, animals, and other vehicles. Summon is unlikely to operate as intended, and/or may abort, in the following types of situations:

- There is an accessory such as a roof rack, trailer hitch, body kit, etc. mounted on or otherwise attached to Model Y. When Summon is steering Model Y to avoid detected obstacles, it does not account for accessories that extend beyond Model Y.
- A raised edge is detected. Summon does not move Model Y over steep edges.
- Weather conditions (heavy rain, snow, fog, or extremely hot or cold temperatures) are interfering with camera operation.
- Model Y is in Trailer Mode or an accessory is attached.
- GPS data is unavailable or there is poor cellular coverage.
- The driving surface is unpaved.
- The driving path has a slope of greater than 20%. Summon is designed to operate only on flat or gently sloped roads (up to 20% grade). When Summon is moving Model Y on a grade higher than 10%, vehicle speed is limted to 3 mph (5 km/h). Summon may not operate at grades above 10% if conditions are poor.

WARNING: The list above does not represent an exhaustive list of situations that may interfere with proper operation of Summon. It is the driver's responsibility to remain in control of Model Y at all times. Pay close attention whenever Summon is actively moving Model Y and stay prepared to take immediate action. Failure to do so can result in serious property damage, injury, or death.

NOTE: Ensure all cameras are clean and free of obstructions before each drive and before using the features described in this topic (see Cleaning a Camera on page 208). Dirty cameras and sensors (if equipped), as well as environmental conditions such as rain and faded lane markings, can affect performance. If a camera is obstructed or blinded, Model Y displays a message on the touchscreen and Self-Driving features may not be available.

WARNING: Summon is a supervised feature. You must continually monitor the vehicle and its surroundings and stay prepared to take immediate action at any time. It is the driver's responsibility to use Actually Smart Summon and Dumb Summon safely, responsibly, and as intended. Summon is designed and intended for use only on parking lots and driveways located on private property where the surrounding area is familiar and predictable. Do not use Actually Smart Summon on public roads. Only use Dumb Summon on public roads with extreme caution. It is your responsibility to familiarize yourself with the limitations of Summon (see Limitations and Warnings on page 134).

WARNING: Actually Smart Summon can abort at any time. If Actually Smart Summon aborts for a safety-related reason, Model Y may stop suddenly or without warning.

WARNING: Always anticipate when you need to stop Model Y. Depending on the quality of the connectivity between the phone and Model Y, there may be a slight delay between when you release the button and when the car stops.

WARNING: Use extreme caution when using Summon in environments where movement of obstacles can be unpredictable. For example, where people, children or animals are present.

WARNING: Actually Smart Summon may not stop for all objects (especially very low objects such as some curbs, or very high objects such as a shelf) and may not react to all oncoming or side traffic. Model Y cannot detect obstacles that are located in the cameras' blind spots near the vehicle's front fenders. Pay attention and be ready to stop Model Y at all times by releasing the button on the mobile app.

WARNING: Summon may not operate as intended when there is an accessory such as a roof rack, trailer hitch, body kit, etc. mounted on or otherwise attached to Model Y. When Summon is steering Model Y to avoid detected obstacles, it does not account for accessories that extend beyond Model Y. Pay attention and be ready to stop Model Y at all times by releasing the button on the mobile app.

WARNING: Model Y cannot detect obstacles that are located in the cameras' blind spots near the vehicle's front fenders, are very narrow, or are hanging from a ceiling (for example, bicycles). In addition, many unforeseen circumstances can impair Summon's ability to move in or out of a parking space and, as a result, Summon may not move Model Y appropriately. Therefore, you must continually monitor the vehicle's movement and its surroundings and remain prepared to stop Model Y at any time.

Self-Driving Limitations and Warnings

This topic includes warnings, cautions, and limitations pertaining to the following Self-Driving features.

NOTE: *Depending on market region, vehicle configuration, options purchased, Self-Driving hardware, and software version, your vehicle may not be equipped with these features, or these features may not operate exactly as described.*

- Traffic-Aware Cruise Control on page 112
- Autosteer on page 115
- Navigate on Autosteer on page 138
- Full Self-Driving (Supervised) on page 105
- Autopark on page 139

Ensure all cameras are clean and free of obstructions before each drive and before using the features described in this topic (see Cleaning a Camera on page 208). Dirty cameras and sensors (if equipped), as well as environmental conditions such as rain and faded lane markings, can affect performance. If a camera is obstructed or blinded, Model Y displays a message on the touchscreen and Self-Driving features may not be available.

Before you can use Self-Driving features, and after some Service visits, you must drive a short distance to calibrate cameras. For more information, see Drive to Calibrate Cameras on page 20.

In addition, these features may not work as intended when:

- The road has sharp curves or significant changes in elevation.
- Road signs and signals are unclear, ambiguous, or poorly maintained.
- Visibility is poor (due to heavy rain, snow, hail, etc. or poorly lit roadways at night)
- You are driving in a tunnel or next to a highway divider that interferes with the view of the camera(s)
- Bright light (such as from oncoming headlights or direct sunlight) interferes with the view of the camera(s).

The list above does not represent an exhaustive list of situations that may interfere with proper operation of Self-Driving features. For more information, see Limitations and Warnings on page 136.

WARNING: Read the following warnings and limitations carefully before using Self-Driving. Failure to follow all warnings and instructions can result in property damage, serious injury, or death.

Traffic-Aware Cruise Control

While using Traffic-Aware Cruise Control, **it is the driver's responsibility to stay alert, drive safely, and be in control of the vehicle at all times.** Always keep your eyes on the road when driving and be prepared to take corrective action as needed.

In addition, it is the driver's responsibility to cruise at a safe speed and maintain a safe following distance based on road conditions and applicable speed limits. Be aware of the following limitations while Traffic-Aware Cruise Control is active.

- There may be situations where the cruising speed may not change when the speed limit changes.
- Traffic-Aware Cruise Control does not adapt driving speed based on road and driving conditions. Do not use Traffic-Aware Cruise Control on winding roads with sharp curves, on icy or slippery road surfaces, or when weather conditions (such as heavy rain, snow, fog, etc.) make it inappropriate to drive at a consistent speed.
- Do not rely on Traffic-Aware Cruise Control to maintain an accurate or appropriate following distance.
- Traffic-Aware Cruise Control may be unable to provide adequate speed control because of limited braking capability and hills. It can also misjudge the distance from a vehicle ahead. Driving downhill can increase driving speed, causing Model Y to exceed your set speed (and potentially the road's speed limit).
- Traffic-Aware Cruise Control may occasionally cause Model Y to brake when not required or when you are not expecting it. This can be caused by closely following a vehicle ahead, detecting vehicles or objects in adjacent lanes (especially on curves), etc.
- Due to limitations inherent in the onboard GPS (Global Positioning System), you may experience situations in which Model Y slows down, especially near exits or off-ramps where a curve is detected and/or you are navigating to a destination and not following the route.
- In some cases (such as having insufficient data), Traffic-Aware Cruise Control may not automatically reduce the set speed on the highway interchange or off-ramp.
- Traffic-Aware Cruise Control may not detect all objects and, especially when cruising over 50 mph (80 km/h), may not brake/decelerate when a vehicle or object is only partially in the driving lane or when a vehicle you are following moves out of your driving path and a stationary or slow-moving vehicle or object is in front of you.
- Traffic-Aware Cruise Control may react to vehicles or objects that either do not exist, or are not in your lane of travel, causing Model Y to slow down unnecessarily or inappropriately.

WARNING: Traffic-Aware Cruise Control is particularly unlikely to operate as intended in the following types of situations:

- The road has sharp curves or significant changes in elevation.
- Road signs and signals are unclear, ambiguous, or poorly maintained.
- Visibility is poor (due to heavy rain, snow, hail, etc. or poorly lit roadways at night)
- You are driving in a tunnel or next to a highway divider that interferes with the view of the camera(s)
- Bright light (such as from oncoming headlights or direct sunlight) interferes with the view of the camera(s).

WARNING: The list above does not represent an exhaustive list of situations that may interfere with proper operation of Traffic-Aware Cruise Control. Traffic-Aware Cruise Control can cancel unexpectedly at any time for unforeseen reasons. Always watch the road in front of you and stay prepared to take appropriate action. It is the driver's responsibility to be in control of Model Y at all times.

WARNING: Traffic-Aware Cruise Control is designed for your driving comfort and convenience and is not a collision warning or avoidance system. Never depend on Traffic-Aware Cruise Control to adequately slow down Model Y. Always watch the road in front of you and be prepared to take corrective action at all times. Failure to do so can result in serious injury or death.

WARNING: Although Traffic-Aware Cruise Control is capable of detecting pedestrians and cyclists, never depend on Traffic-Aware Cruise Control to adequately slow Model Y down for them. Failure to do so can result in serious injury or death.

Autosteer

WARNING: Autosteer is a hands-on feature. Keep your hands on the steering wheel at all times, be mindful of road conditions and surrounding traffic, and always be prepared to take immediate action. Failure to follow these instructions could cause damage, serious injury or death.

WARNING: Autosteer is intended for use on controlled-access highways with a fully attentive driver. Do not use Autosteer in construction zones, or in areas where bicyclists or pedestrians may be present.

WARNING: Never depend on Autosteer to determine an appropriate driving path.

CAUTION: Autosteer and its associated functions are particularly unlikely to operate as intended when:

- Autosteer is unable to accurately determine lane markings. For example, lane markings are excessively worn, have visible previous markings, have been adjusted due to road construction, are changing quickly (lanes branching off, crossing over, or merging), objects or landscape features are casting strong shadows on the lane markings, or the road surface contains pavement seams or other high-contrast lines.
- Visibility is poor (heavy rain, snow, fog, etc.) or weather conditions are interfering with sensor operation.
- A camera(s) or sensor(s) is obstructed, covered, or damaged.
- Driving on hills.
- Approaching a toll booth.
- Driving on a road that has sharp curves or is excessively rough.
- Bright light (such as direct sunlight) is interfering with the view of the camera(s).
- The sensors (if equipped) are affected by other electrical equipment or devices that generate ultrasonic waves.
- A vehicle is detected in your blind spot when you engage the turn signal.
- Model Y is being driven very close to a vehicle in front of it, which is blocking the view of the camera(s).

WARNING: Many unforeseen circumstances can impair the operation of Autosteer. Always keep this in mind and remember that as a result, Autosteer may not steer Model Y appropriately. Always drive attentively and be prepared to take immediate action.

WARNING: Autosteer is not designed to, and will not, steer Model Y around objects partially in a driving lane and in some cases, may not stop for objects that are completely blocking the driving lane. Always watch the road in front of you and stay prepared to take immediate action. It is the driver's responsibility to be in control of Model Y at all times.

Auto Lane Change

CAUTION: When changing lanes using Auto Lane Change, It is the driver's responsibility to determine whether a lane change is safe and appropriate. Therefore, before initiating a lane change, always check blind spots, lane markings, and the surrounding roadway to confirm it is safe and appropriate to move into the target lane.

CAUTION: Be aware of the following limitations while using Auto Lane Change.

- Never depend on Auto Lane Change to determine an appropriate driving path. Drive attentively by watching the road and traffic ahead of you, checking the surrounding area, and monitoring the touchscreen for warnings. Always be prepared to take immediate action.
- Do not use Auto Lane Change on roads where traffic conditions are constantly changing and where bicycles and pedestrians are present.
- The performance of Auto Lane Change depends on the ability of the camera(s) to recognize lane markings.
- Do not use Auto Lane Change on winding roads with sharp curves, on icy or slippery roads, or when weather conditions (such as heavy rain, snow, fog, etc.) may be obstructing the view from the camera(s) or sensors (if equipped).
- Overtake Acceleration can cancel for many unforeseen reasons in addition to those listed above (for example, lack of GPS data). Stay alert and never depend on Overtake Acceleration to increase your driving speed.
- Overtake Acceleration increases your driving speed whenever the appropriate turn signal is engaged, and accelerates Model Y closer to the vehicle ahead. Although Traffic-Aware Cruise Control continues to maintain distance from the vehicle ahead, it is important to be aware that your selected following distance is reduced when Overtake Acceleration is active, particularly in cases where it may not be your intention to overtake the vehicle you are following.

Stop Light and Stop Sign Warning

WARNING: Stop Light and Stop Sign Warning requires on-board maps to know that a particular stop light or stop sign exists at a location. In some cases, map data is inaccurate or outdated and may not include all stop lights or stop signs. Therefore, Stop Light and Stop Sign Warning may not detect all stop lights and stop signs.

WARNING: The Stop Light and Stop Sign Warning feature does not apply the brakes or decelerate Model Y and may not detect all stop lights and stop signs. Stop Light and Stop Sign Warning is designed for guidance purposes only and is not a substitute for attentive driving and sound judgment. Keep your eyes on the road when driving and never depend on Stop Light and Stop Sign Warning to warn you of a stop light or stop sign.

WARNING: Stop Light and Stop Sign Warning is designed to warn you only when approaching a visible red stop sign, solid red or later portion of a yellow traffic light. It may not warn you of intersections with flashing lights and it does not warn you of yield signs or temporary stop and yield signs (such as those used in construction areas). Additionally, Stop Light and Stop Sign Warning does not warn you of approaching stop lights or stop signs when you are pressing the accelerator pedal or brake pedal (which disables Autosteer).

Navigate on Autosteer

WARNING: Never depend on Navigate on Autosteer to determine an appropriate lane at an off-ramp. Stay alert and perform visual checks to ensure that the driving lane is safe and appropriate.

WARNING: If you turn off **Require Lane Change Confirmation**, Navigate on Autosteer notifies you of upcoming lane changes and off-ramps, but it remains your responsibility to monitor the environment and maintain control of Model Y at all times. Lane changes can occur quickly and suddenly. Always keep your hands on the wheel and your eyes on the driving path in front of you.

WARNING: Navigate on Autosteer does not make driving autonomous. You must pay attention to the road, keep your hands on the steering wheel at all times, and remain aware of your navigation route.

WARNING: As is the case with normal driving, be extra careful around blind corners, interchanges, and on-ramps and off-ramps - obstacles can appear quickly and at any time.

WARNING: Navigate on Autosteer may not recognize or detect oncoming vehicles, stationary objects, and special-use lanes such as those used exclusively for bikes, carpools, emergency vehicles, etc. Remain alert at all times and be prepared to take immediate action. Failure to do so can cause damage, injury or death.

Autopark

CAUTION: Autopark's performance depends on the ability of the cameras and sensors (if equipped) to determine the vehicle's proximity to curbs, objects, and other vehicles. Be aware of the following warnings before and while using Autopark:

- Do not use Autopark if anything, such as a ball hitch, bike rack, or trailer, is attached to the tow hitch. Autopark may not stop for hitches when parking between or in front of other vehicles.
- Never depend on Autopark to find a parking space that is legal, suitable, and safe. Autopark may not always detect objects in the parking space. Always perform visual checks to confirm that a parking space is appropriate and safe.
- When Autopark is actively steering Model Y, the steering wheel moves in accordance with Autopark's adjustments. Do not interfere with the movement of the steering wheel. Doing so cancels Autopark.
- During the parking sequence, continually check your surroundings. Be prepared to apply the brakes to avoid vehicles, pedestrians, or objects.
- When Autopark is active, monitor the touchscreen to ensure that you are aware of the instructions that Autopark is providing.

CAUTION:

Autopark is particularly unlikely to operate as intended in these situations:

- The road is sloped. Autopark is designed to operate on flat roads only.
- Visibility is poor (due to heavy rain, snow, fog, etc.).
- The curb is constructed of material other than stone, or the curb cannot be detected.
- The target parking space is directly adjacent to a wall or pillar (for example, the last parking space of a row in an underground parking structure).
- One or more of the sensors (if equipped) or cameras is damaged, dirty, or obstructed (such as by mud, ice, or snow, or by a vehicle bra, excessive paint, or adhesive products such as wraps, stickers, rubber coating, etc.).
- The sensors (if equipped) are affected by other electrical equipment or electrical interference.

WARNING: Many unforeseen circumstances can impair Autopark's ability to park Model Y. Keep this in mind and remember that as a result, Autopark may not steer Model Y appropriately. Pay attention when parking Model Y and stay prepared to immediately take control.